MATEO CASTANO-YEPES

Kitchener, ON | (519) 589-7029 | m.castanoyepes@hotmail.com | www.linkedin.com/in/mateo-cy

EDUCATION

University of Toronto | 3.7/4.0 cGPA — **Toronto, ON**
B.A.Sc. in Mechanical Engineering + PEY Co-op — *Class of 2029 + PEY*

- $15,000 Scholarship Recipient, Blue Sky Solar Racing Team Member, George Brown Introductory Machining Course.

WORK EXPERIENCE

Mechanical Engineer Intern (Part-Time) — **San Diego, CA (Remote)**
Space Copy — *June 2026 – September 2026*

- Aided in the development of autonomous ISRU **3D printing** systems for lunar missions under a phase II NASA contract.
- Owned the design of adjustable build plate heating and levelling sub-system designs, modelled with **SolidWorks**.
- Designed a robotic arm attachment to release printed parts from the build plate minimizing, modelled with **SolidWorks.**
- Researched different build plate material options with an emphasis on effectively 3D printing lunar regolith simulants.
- Fully rendered a digital twin of the 3D printer using **Blender** to accurately represent different materials and lighting.

Health & Safety Coordinator — **Waterloo, ON**
BrickWorks Academy — *July 2026 –September 2026*

- Implemented safety procedures, first aid and daily cleaning to maintain camper safety across 4 STEM summer camps.
- Instructed hundreds of campers on using **CAD** software, **3D printing**, **DFM**, and building/coding with LEGO Robotics.

Precast Manufacturing Engineer Intern — **Melbourne, AUS**
SH Engineering Pty Ltd — *May 2026 – July 2026*

- Led the integration of a hazardous gas alarm system with PLC infrastructure, modelled in **AutoCAD** and **SolidWorks**.
- Verified structural integrity and quality through inspections of 200+ precast segments for two Australian tunnel projects.
- Increased carousel production by **11%** by logging 5+ hours of production cycle times across the line **(Excel)**, identifying main bottlenecks and producing a report with recommendations on how to distribute resources to decrease cycle times.
- Aided in the development of two 15+ page engineering proposals identifying various areas of improvement on the site to increase production by **33%** and increase segment storage capacity by **15,000+** to prevent overflow from production.
- Compiled daily shift report data into an organized production tracker, automating weekly average calculations in **Excel.**

PROJECT/EXTRACURRICULAR EXPERIENCE

Structural/Fabrication Team Member — **Toronto, ON**
UofT Blue Sky Solar Racing — *September 2026 – Present*

- Designed the rollcage of the driver compartment using **CATIA,** adhering to American Solar Challenge Guidelines.
- Iterated over multiple rollcage geometries to minimize weight and allow for optimal aerodynamic performance.
- Verified rollcage structural integrity to ensure driver safety using **Ansys FEA** simulations and prototype testing.
- Manufactured various structural and aerodynamic carbon fiber parts using hand layup and resin infusion techniques.

Research Lead and Developer — **Toronto, ON**
TriggerPredict (For Mount Sinai Fertility Clinic) — *January 2026 – April 2026*

- Coordinated research objectives and tasks within a team of five on In-Vitro Fertilization trigger timing and related ML models with the goal of creating a highly predictive, user-friendly algorithm to predict optimal IVF trigger timing.
- Formulated **124** conceptual designs from the client statement, documented in three technical reports totaling 97 pages.
- Developed the TriggerPredict app end to end using **Python** and Typescript, scoring a **8.2/10** System Usability Score.

SKILLS

Hardware: Machining, Lathe, Mill, 3D Printing, Soldering, Circuit Design, Woodworking, Prototyping, Material Testing
CAD Software: SolidWorks **(CSWA)**, CATIA, Fusion 360, Onshape, AutoCAD, Blender (Rendering)
Analysis & Programming: Ansys Mechanical (FEA), Python, MATLAB, LabVIEW, Excel (Data Analysis/Automation)
Engineering Tools: GD&T, DFM, Jira, Root Cause Analysis, FMEA, Minitab Statistical Software, Six Sigma White Belt
Soft Skills: Problem Solving, Leadership, Technical Writing, Project Management, Communication, Teamwork, Spanish